Worksheet: Choosing Price and Quantity

ECON 201

Definitions

- **Profit**: Profit $= R - C(Q)$, where revenue is $R = P \times Q$.
- **Marginal cost**: the increase in total cost when one additional unit is produced.
- **Marginal revenue**: the increase in revenue when one additional unit is sold. To sell one more, the firm lowers its price on every unit, so
  - MR $=$ price of the extra unit $-$ the loss on the units it was already selling.
- **Profit-maximizing quantity**: where MR $=$ MC.
  - While MR $>$ MC, one more unit adds to profit. Once MR $<$ MC, one more unit lowers it.

Boba Break

Boba Break is a stall at the Tuesday farmers market on campus. It pays \$100 a day for the stall and \$4 to make each cup, so $C(Q) = 100 + 4Q$. Its demand curve is $P = 12 - Q/10$: the more cups Boba Break wants to sell, the lower the price it has to charge.

1. Marginal cost

How much does one more cup add to Boba Break's cost? Does it matter how many cups it is already selling?

2. Marginal revenue at 30 cups

(a) What price does Boba Break have to charge to sell 30 cups? What about 31 cups?

(b) What is its revenue at 30 cups, and at 31 cups?

(c) What is the marginal revenue? Should Boba Break sell the 31st cup?

3. Marginal revenue at 50 cups

Now Boba Break sells 50 cups at \$7, and a 51st cup would sell for \$6.90. What is the marginal revenue? Should it sell the 51st cup?

4. The best quantity

Somewhere between 30 and 50 cups, marginal revenue falls below \$4. Where should Boba Break stop, and what price does it charge?